Proficient in Civil 3D (80 Hours)				
S.NO	**Session**	**Topics**	**Type**	**Duration (Hours)**
1	Session 1	Introduction Land Survey Land Development Transportation Design AutoCAD Civil 3D Working Environment Work Space Tool Space Object Object Style Label Style Creating Template Creating Reports	Theory/Demo	2.0 Hrs
2	Session 2	Exercises included for Practice	Practical	2.0 Hrs
3	Session 3	Working with Point Data Creating and Editing Points Miscellaneous Intersection	Theory/Demo	2.0 Hrs
4	Session 4	Exercises included for Practice	Practical	2.0 Hrs
5	Session 5	Working with Point Data Creating and Editing Points Slope Interpolate Point Styles Point Group Import/Export Points	Theory/Demo	2.0 Hrs
6	Session 6	Exercises included for Practice	Practical	2.0 Hrs
7	Session 7	Surface How to create a surface based on Points Modifying Surface Surface Styles Surface Label Styles Create Points from Surface	Theory/Demo	2.0 Hrs
8	Session 8	Exercises included for Practice	Practical	2.0 Hrs
9	Session 9	**Survey:** Total Station Creating Survey Database Creating Survey Network Creating Figure Styles and Prefix Importing Field Book Working with Survey Data Creating Surface	Theory/Demo	2.0 Hrs
10	Session 10	Exercises included for Practice	Practical	2.0 Hrs
11	Session 11	LiDAR Survey Introduction to LiDAR How to Import LiDAR Survey Data Creating Surface Working with DEM files	Theory/Demo	2.0 Hrs
12	Session 12	Exercises included for Practice	Practical	2.0 Hrs
13	Session 13	Surface Analysis Direction Elevation Slope Slope Arrow Water Shed Water Drop Catchment Area Legend Table	Theory/Demo	2.0 Hrs
14	Session 14	Exercises included for Practice	Practical	2.0 Hrs
		Earthwork Calculation Create surface by Breaklines		

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15	Session 15	Contours Drawing Objects Alignment Creation Horizontal Profile View generation Customising Profile Views Create a Volume Surface Volume Calculation	Theory/Demo	2.0 Hrs
16	Session 16	Exercises included for Practice	Practical	2.0 Hrs
17	Session 17	Grading Xref Combining Surface from multiple files Global Coordinates	Theory/Demo	2.0 Hrs
18	Session 18	Exercises included for Practice	Practical	2.0 Hrs
19	Session 19	Parcel Site Creating Parcels From Object By Layout Creating Right Of Way Editing Parcels Labelling Parcels Creating Tables	Theory/Demo	2.0 Hrs
20	Session 20	Exercises included for Practice	Practical	2.0 Hrs
21	Session 21	Project: Land Development	Theory/Demo	2.0 Hrs
22	Session 22	Exercises included for Practice	Practical	2.0 Hrs
23	Session 23	Alignment Design as per IRC Design Criteria Editor Alignment from Polyline Alignment Style Alignment Label Style Design Criteria Design Check Set Alignment by Layout Edit Alignment Geometry Tables Best Fit Alignment Create Offset Alignment Create Widening	Theory/Demo	2.0 Hrs
24	Session 24	Exercises included for Practice	Practical	2.0 Hrs
25	Session 25	Profile Design as per IRC Profile by Layout Multiple Profile Views Super Imposed Profile Edit Profile Geometry Design Criteria Editor Profile View Labels	Theory/Demo	2.0 Hrs
26	Session 26	Exercises included for Practice	Practical	2.0 Hrs
27	Session 27	Assembly Create Assembly from Library Create Sub-assembly from polyline Assembly Offset	Theory/Demo	2.0 Hrs
28	Session 28	Exercises included for Practice	Practical	2.0 Hrs
29	Session 29	Corridor and Corridor Surface Create Corridor View/Edit Corridor Sections Super Elevation Corridor Surface Corridor Surface Boundaries	Theory/Demo	2.0 Hrs
30	Session 30	Exercises included for Practice	Practical	2.0 Hrs
31	Session 31	Junction Design Manual Junction Design (based on versions before 2010) Automatic Junction Design Customise the Assemblies	Theory/Demo	2.0 Hrs

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		Corridor Surface		
32	Session 32	Exercises included for Practice	Practical	2.0 Hrs
33	Session 33	Roundabout Design Add Approach Add Turn Slip Land Create Corridors Corridor Surface Grading Feature Lines Grading Surface Label	Theory/Demo	2.0 Hrs
34	Session 34	Exercises included for Practice	Practical	2.0 Hrs
35	Session 35	Quantity Takeoff Sample Lines Section Views Compute Materials Earthwork Material Quantity Takeoff Creating New Criteria	Theory/Demo	2.0 Hrs
36	Session 36	Exercises included for Practice	Practical	2.0 Hrs
37	Session 37	Pipe Network Parts List Pipe Network by Layout Pipe Network from Object Edit Pipe Network Draw Parts in Profile View Create and Apply Rules Pipe Network Labels Pipe Network Tables	Theory/Demo	2.0 Hrs
38	Session 38	Exercises included for Practice	Practical	2.0 Hrs
39	Session 39	**Project:** Transportation Design	Theory/Demo	2.0 Hrs
40	Session 40	Exercises included for Practice	Practical	2.0 Hrs